

Course: B. Tech.

Branch: Group A/ Group B

Semester: I

Subject Code & Name: BTES105 Energy and Environment Engg.

Max Marks: 60

Date: 19/8/2023

Duration: 3 Hrs.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve any two of the following.		12
A) Explain the working of a Steam power plant with neat diagram. Write down the advantages and disadvantages of Steam Power plant.	CO1	6
B) Explain how electricity is generated from a nuclear reactor? What is the major hazard of nuclear power generation?	CO1	6
C) Write the correct type of energy produced by the following power plants: a) Tarapur power plant b) Koyna Power plant c) Uran Power Plant d) Durgapur Power Plant And write down the advantages of Hydro-power plant.	CO1	6
Q.2 Solve Any Two of the following.		12
A) How are the windmills classified? With a neat diagram, explain how wind energy can be converted into electrical energy	CO2	6
B) What is Bio-mass? Write construction and working of bio-gas plant, with a neat diagram. Also write down the advantages of it.	CO2	6
C) What are the main types of OTEC power plants? Describe the working of open cycle OTEC with a neat diagram.	CO2	6
Q. 3 Solve Any Two of the following.		12
A) What do you mean by energy conservation? Explain the measures to be taken to reduce the energy conservation in domestic activities. List any four measures.	CO2	6
B) What do you understand by maximum energy efficiency in context with energy conservation principle? Discuss with a suitable example.	CO1	6
C) Write down methods of energy conservation in refrigeration and AC.	CO2	6
Q.4 Solve Any Two of the following.		12
A) Define Indoor Air Pollution. Write down the different sources of indoor air pollution.	CO3	6
B) Describe the measures to be taken to control the noise pollution.	CO1	6

C)	Define Soil Pollution. Write down various causes of soil pollution. What are the effects of Soil pollution?	CO3	6
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Q. 5	Solve any four of the following.		12
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A)	What are the main causes of water pollution?	CO3	3
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B)	Explain the concept of “Acid rain”.	CO3	3
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C)	What are the various methods of safe disposal of solid wastes?	CO3	3
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D)	Write down the sources of noise pollution.	CO3	3
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E)	What is radioactive pollution? What are its effects?	CO3	3
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F)	What do you mean by “Thermal Pollution”? What are its effects?	CO3	3
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